

Boundary value problems and integral operators

a calculation for weeks 2 & 3 (*version 2*)

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Outline

- 1 a classic boundary value problem
- 2 integral operators
- 3 compact subsets of $C([0, 1])$
- 4 bounded and compact operators
- 5 other views of the boundary value problem

second-order BVP (“2-point problem”)

suppose $\alpha, \beta \in \mathbb{R}$ and $\phi \in C([0, 1])$ are given. solve for $u(x)$:

$$-u''(x) = \phi(x) \quad \text{for all } x \in (0, 1)$$

$$u(0) = \alpha$$

$$u(1) = \beta$$

- picture of data ϕ (left) and solution u (right):
- we will see this is *solvable by hand*, at least as an integral of the data $\phi(x)$, but this is mildly difficult, so we warm-up with an easier problem

second-order IVP

suppose $\alpha, \beta \in \mathbb{R}$ and $\phi \in C([0, 1])$ are given. solve for $u(x)$:

$$-u''(x) = \phi(x) \quad \text{for all } x \in (0, 1)$$

$$u(0) = \alpha$$

$$u'(0) = \beta$$

- picture:

solution of IVP

apply fundamental theorem of calculus twice:

$$-u''(x) = \phi(x)$$

solution of BVP

apply fundamental theorem of calculus twice:

← it will be more work

$$-u''(x) = \phi(x)$$

result:

$$u(x) = \alpha + (\beta - \alpha)x + \int_0^1 k(x, t)\phi(t) dt$$

where k is a symmetric *kernel*:

$$k(x, t) = \begin{cases} t(1-x), & 0 \leq t \leq x \leq 1 \\ x(1-t), & 0 \leq x \leq t \leq 1 \end{cases} = \min\{x, t\} - xt$$

$k(x, t)$ for the BVP

picture of $k(x, t)$:

- restrict to the $\alpha = \beta = 0$ special case

2-point BVP

the solution of

$$-u'' = \phi, \quad u(0) = u(1) = 0$$

is

$$u(x) = \int_0^1 k(x, t) \phi(t) dt \quad \text{where } k(x, t) = \min\{x, t\} - xt$$

- u is the output of a linear *integral operator*

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1. a classic boundary value problem
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Definition

given a suitable *kernel* $k(x, t)$, the *integral operator* $L : C([0, 1]) \rightarrow C([0, 1])$ is

$$(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$$

- L is linear:
- is L a matrix?

wait ... does $\phi \in C([0, 1]) \implies L\phi \in C([0, 1])$?

Definition

given a **suitable** kernel $k(x, t)$, the *integral operator* $L : C([0, 1]) \rightarrow C([0, 1])$ is

$$(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$$

- what does “suitable” mean?
- is $L\phi$ even well-defined? is $L\phi \in C([0, 1])$?
- for $\phi \in C([0, 1])$, we see that $(L\phi)(x)$ is well-defined if $k(x, t)$ is (Riemann) integrable in t for every x
- what conditions on $k(x, t)$ guarantee that

$$\phi \in C([0, 1]) \implies L\phi \in C([0, 1])?$$

- does $k(x, t)$ itself need to be continuous?

example: the antiderivative (as an integral operator)

Definition

for fixed $a \in [0, 1]$, an *antiderivative operator* $A : C([0, 1]) \rightarrow C([0, 1])$ is

$$(A\phi)(x) = \int_a^x \phi(t) dt$$

- the FTC shows this is an antiderivative:
- these integrals with variable limits give *Volterra integral operators*, whereas L on the previous slide is a *Fredholm integral operator*
 - look it up?

kernel of the antiderivative operator

- can the antiderivative operator $(A\phi)(x) = \int_0^x \phi(t) dt$ be put into the kernel form $(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$?
- yes, and $k(x, t)$ is discontinuous (picture at right):

$$k(x, t) = \begin{cases} 1, & t \leq x \\ 0, & t > x \end{cases} = \mathbb{1}_{[0, x]}(t)$$

- compute:

$$(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt = \int_0^1 \mathbb{1}_{[0, x]}(t)\phi(t) dt = \int_0^x \phi(t) dt = (A\phi)(x)$$

Lipschitz kernels

- regarding what conditions on $k(x, t)$ imply $L : C([0, 1]) \rightarrow C([0, 1])$, here is a sufficient condition

Definition

a function $k : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ is *Lipschitz in x* if there is $C \geq 0$ so that for all $x, y \in [0, 1]$ and $t \in [0, 1]$, $|k(x, t) - k(y, t)| \leq C|x - y|$

Lemma

if $k(x, t)$ is Lipschitz in x , and if $(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$ for $\phi \in C([0, 1])$, then $L\phi \in C([0, 1])$

- proof on next slide
- this is *not* a comprehensive theory of integral operators, but it gives me something to prove with elementary tools

Lemma

if $k(x, t)$ is Lipschitz in x , and if $(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$ for $\phi \in C([0, 1])$, then $L\phi \in C([0, 1])$

Proof. Let $\phi \in C([0, 1])$, and consider the function $L\phi$ at a point $x \in [0, 1]$. Suppose $\epsilon > 0$. Let $\delta = \epsilon/(C\|\phi\|_\infty)$. If $y \in [0, 1]$ and $|x - y| < \delta$ then by the triangle inequality for integrals, and the definition of Lipschitz,

$$\begin{aligned} |(L\phi)(x) - (L\phi)(y)| &= \left| \int_0^1 k(x, t)\phi(t) dt - \int_0^1 k(y, t)\phi(t) dt \right| \\ &\leq \int_0^1 |k(x, t) - k(y, t)| |\phi(t)| dt \leq \int_0^1 C|x - y| |\phi(t)| dt \\ &\leq \int_0^1 C|x - y| \|\phi\|_\infty dt = C\|\phi\|_\infty |x - y| < C\|\phi\|_\infty \delta = \epsilon \end{aligned}$$

□

recall: solution of 2-point BVP

the solution of

$$-u'' = \phi, \quad u(0) = u(1) = 0$$

is $u(x) = \int_0^1 k(x, t) \phi(t) dt$ where

$$k(x, t) = \begin{cases} t(1-x), & t \leq x \\ x(1-t), & x \leq t \end{cases} = \min\{x, t\} - xt$$

Corollary

since the above kernel is Lipschitz in x , with $C = 0.5$, it follows that $u \in C([0, 1])$ if $\phi \in C([0, 1])$

- but we already knew that *this* L sends $C([0, 1])$ to itself ... why?
- in fact: if $\phi \in C([0, 1])$ then $u(x)$ has two derivatives

bounded kernel is not enough

- Q. is integrability and boundedness of k sufficient for $L : C([0, 1]) \rightarrow C([0, 1])$?
- A. *no.* there exist integrable, bounded, and discontinuous $k(x, t)$ so that $(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$ is not continuous

example

$$k(x, t) = \begin{cases} 1, & x \leq 0.5 \\ 0, & x > 0.5 \end{cases}$$

and $\phi(x) = 1$ gives

$$(L\phi)(x) =$$

- regarding necessary and sufficient conditions, I do not know!
- feel free to do research and try it yourself

pictures of three kernels

on t, x axes, sketch contour maps?:

$$k_1(x, t) = \begin{cases} t(1-x), & t \leq x \\ x(1-t), & x \leq t \end{cases}$$

$$k_2(x, t) = \begin{cases} 1, & t \leq x \\ 0, & t > x \end{cases}$$

$$k_3(x, t) = \begin{cases} 1, & x \leq 0.5 \\ 0, & x > 0.5 \end{cases}$$

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$C([0, 1])$, $\|\cdot\|_\infty$ is a normed vector space

- recall stuff from Chapter 1 of Saxe, *Beginning Functional Analysis*

Definition

for $f \in C([0, 1])$, let
$$\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)| = \max_{x \in [0, 1]} |f(x)|$$

- picture:
- if $V = C([0, 1])$ then $(V, \|\cdot\|_\infty)$ is a normed vector space
 - there are four properties to check: (i), (ii), (iii), (iv)
 - recall: $d(f, g) = \|f - g\|_\infty$ makes (V, d) a metric space
- recall: convergence in $\|\cdot\|_\infty$ is *uniform* convergence

compact sets

- now Chapter 2 of Saxe, *Beginning Functional Analysis*
- we assume $(V, \|\cdot\|)$ is a normed vector space
- recall: a subset $S \subset V$ is *open* if $\forall x \in S \exists r > 0$ so that $B_r(x) \subset S$
 - $B_r(x) = \{y \in V : \|y - x\| < r\}$ is an *open ball*

Definition

$\mathcal{C} = \{S_\lambda\}_{\lambda \in I}$ is an *open cover* of $E \subset V$ if $S_\lambda \subset V$ is open for each $\lambda \in I$ and $E \subset \bigcup_{\lambda \in I} S_\lambda$

Definition

- a subset $E \subset V$ is *compact* if every open cover of E has a finite subcover
- a subset $E \subset V$ is *sequentially compact* if every sequence in E contains a convergent subsequence
- fact: a finite set is compact

Theorem 2.4

$E \subset V$ is compact if and only if it is sequentially compact

the unit ball in $V = C([0, 1])$ is not compact

Theorem 2.3

if $E \subset V$ is compact then it is closed

Theorem

the closed unit ball $B_1(0) = \{y \in V = C([0, 1]) : \|y\|_\infty \leq 1\}$ is not compact

Proof. The hat functions $\psi_n(x)$, $n \in \mathbb{N}$, drawn below, are continuous, and each has norm $\|\psi_n\|_\infty = 1$, thus $\psi_n \in B_1(0)$. Also, for any $n \neq m$, $\|\psi_n - \psi_m\|_\infty = 1$. Any subsequence (ψ_{n_k}) of the sequence (ψ_n) would not converge because again $\|\psi_{n_j} - \psi_{n_k}\|_\infty = 1$ for $j \neq k$. Thus $B_1(0)$ is not compact. $\square$

- *Corollary.* if $E \subset V = C([0, 1])$ contains a ball of positive radius $B_r(x) \subset E$, no matter how small $r > 0$ is, then E is not compact

why would we want sets to be compact?

Extreme Value Theorem (Weierstrass)

if $E \subset V$ is compact and $f : E \rightarrow \mathbb{R}$ is continuous then there exists a maximum and a minimum of f

- here $(V, \|\cdot\|)$ is any normed vector space
- the conclusion is that there are $c, C \in E$ so that

$$f(c) \leq f(x) \leq f(C)$$

for all $x \in E$

- slightly more generally, one can consider continuous $f : M \rightarrow \mathbb{R}$ where (M, d) is any compact metric space
- compact subsets of $V = C([0, 1])$ are a major focus of Chapter 2

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bounded and compact operators

- assume $(V, \|\cdot\|)$ is a normed vector space
- the closed unit ball is $B = \{y \in V : \|y\| \leq 1\}$

Definition

- a linear operator $L : V \rightarrow V$ is *bounded* if there is a constant $C \geq 0$ so that

$$\|Lv\| \leq C\|v\| \quad \text{for all } v \in V$$

- a linear operator $L : V \rightarrow V$ is *compact* if the image of the closed unit ball under L , that is, the set LB , is compact
- we will see that “bounded” is the same as “continuous” for *linear* operators

bounded integral operators on $C([0, 1])$

Theorem

Let $V = C([0, 1])$ with $\|\cdot\|_\infty$ as the norm. Consider the operator $L : V \rightarrow V$,
 $(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$. If $k(x, t)$ is integrable and

$$\sup_{x \in [0, 1]} \int_0^1 |k(x, t)| dt < \infty$$

then L is bounded.

Proof. Suppose $\phi \in V$. Then by the triangle inequality,

$$\begin{aligned} \|L\phi\|_\infty &= \sup_{x \in [0, 1]} |(L\phi)(x)| = \sup_{x \in [0, 1]} \left| \int_0^1 k(x, t)\phi(t) dt \right| \\ &\leq \sup_{x \in [0, 1]} \int_0^1 |k(x, t)| |\phi(t)| dt \leq \sup_{x \in [0, 1]} \int_0^1 |k(x, t)| \|\phi\|_\infty dt \\ &= \left(\sup_{x \in [0, 1]} \int_0^1 |k(x, t)| dt \right) \|\phi\|_\infty \end{aligned}$$

Thus $\|L\phi\|_\infty \leq C\|\phi\|_\infty$ where $C = \sup_{x \in [0, 1]} \int_0^1 |k(x, t)| dt$. □

new idea: equicontinuous sets of functions

Definition (see section 2.1)

a set $E \subset C([a, b])$ is *equicontinuous at* $x \in [a, b]$ if for each $\epsilon > 0$ there is $\delta > 0$ so that $y \in [a, b]$ and $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$ for all $f \in E$

- this is about quantifiers!
- f continuous at $x \in [a, b]$:

$$\forall \epsilon > 0 \exists \delta > 0 \forall y \in [a, b] \quad |x - y| < \delta \implies |f(x) - f(y)| < \epsilon$$

- E equicontinuous at $x \in [a, b]$:

$$\forall \epsilon > 0 \exists \delta > 0 \forall f \in E \forall y \in [a, b] \quad |x - y| < \delta \implies |f(x) - f(y)| < \epsilon$$

Definition

a set $E \subset C([a, b])$ is *equicontinuous* if it is equicontinuous at each $x \in [a, b]$

- more on equicontinuity in Chapter 2 of Saxe, *Beginning Functional Analysis*

new idea: some integral operators are compact

Ascoli-Arzelà Theorem

a set $E \subset C([a, b])$ is compact, in the topology of the $\|\cdot\|_\infty$ norm, if and only if it is closed, bounded, and equicontinuous

- now we can return to the BVP!

Theorem

Consider $B_R(0) \subset V = C([0, 1])$. For $\phi \in B_R(0)$ the solution of

$$-u'' = \phi, \quad u(0) = u(1) = 0$$

is $u(x) = (L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$ where $k(x, t) = \min\{x, t\} - xt$. Then $u \in E$ where $E = LB_R(0)$ is a compact subset of V .

Proof. Show E is closed and bounded. Show E equicontinuous by using the fact that $k(x, t)$ is Lipschitz in x . □

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the Dirac delta “function”

Definition

the *Dirac delta function* on $[0, 1]$, centered at $t \in (0, 1)$, satisfies

$$\delta_t(x) = \begin{cases} 0, & x \neq t \\ \text{positive}, & x = t, \end{cases}$$

and

$$\int_0^1 \delta_t(x) dx = 1$$

- picture:

- the above is a **fake** definition . . . do not believe it
- the Dirac delta “function” is actually a positive *measure* $\delta_t(x) dx$

the kernel for the BVP comes from Dirac delta functions

Theorem

for $t \in (0, 1)$, the solution of

$$-u''(x) = \delta_t(x), \quad u(0) = u(1) = 0$$

is $k(x, t) = \min\{x, t\} - xt$

Proof. Integrate twice, as usual:

$$u'(x) = C - \int_0^x \delta_t(s) ds = C - \mathbb{1}_{(t,1)}(x)$$

$$u(x) = 0 + \int_0^x C - \mathbb{1}_{(t,1)}(s) ds = Cx - \begin{cases} 0, & x < t \\ x - t, & x > t \end{cases}$$

But:

$$0 = u(1) = C - \begin{cases} 0, & 1 < t \\ 1 - t, & 1 > t \end{cases} = C - (1 - t)$$

so $C = 1 - t$ so:

$$u(x) = (1 - t)x - \begin{cases} 0, & x < t \\ x - t, & x > t \end{cases} = \begin{cases} x(1 - t), & x < t \\ t(1 - x), & x > t \end{cases} = k(x, t)$$

□

- this proof is valid if we interpret $\delta_t(x) dx$ as a measure

Corollary

for $t \in (0, 1)$, the solution of

$$-u''(x) = \phi(x), \quad u(0) = u(1) = 0$$

is $u(x) = \int_0^1 k(x, t)\phi(t) dt$ where $k(x, t) = \min\{x, t\} - xt$

“Proof.” The map from ϕ to the solution u is linear: $u = L\phi$. Express ϕ as a linear combination of delta functions:

$$\phi(x) = \int_0^1 \phi(t)\delta_t(x) dt$$

Now apply L to both sides, to write the solution as a linear combination of solutions:

$$u(x) = (L\phi)(x) = \int_0^1 \phi(t)L(\delta_t(x)) dt$$

But from previous slide, $L(\delta_t(x)) = k(x, t)$. Thus

$$u(x) = \int_0^1 \phi(t)k(x, t) dt = \int_0^1 k(x, t)\phi(t) dt \quad \square$$

- this “proof” becomes valid in the theory of *distributions* or *generalized functions*, due to L. Schwartz

approximation of the BVP

- suppose that an integral operator $L : C([0, 1]) \rightarrow C([0, 1])$ has formula

$$(L\phi)(x) = \int_0^1 k(x, t)\phi(t) dt$$

- neither L nor k is a matrix, but could we approximate them by a matrix?
- one answer: finite elements
- picture of a test (hat) function $\phi(x) = \psi_k(x)$ and the output $L\psi_k$:

solution of the BVP by Fourier series

Lemma

for $n \in \mathbb{N}$, the solution of

$$-u''(x) = \sin(n\pi x), \quad u(0) = u(1) = 0$$

is
$$u(x) = \frac{1}{n^2\pi^2} \sin(n\pi x)$$

- the proof is very easy?
- suppose data $\phi(x)$ is written as a Fourier sine series (Chapter 4):

$$\phi(x) = \sum_{n=1}^{\infty} c_n \sin(n\pi x)$$

- then, by linearity of $(\cdot)''$, the solution of the BVP is

$$u(x) = \sum_{n=1}^{\infty} \frac{c_n}{n^2\pi^2} \sin(n\pi x)$$

solution of the BVP by Fourier series 2

- we have $\phi(x) = \sum_{n=1}^{\infty} c_n \sin(n\pi x)$ and $u(x) = \sum_{n=1}^{\infty} \frac{c_n}{n^2\pi^2} \sin(n\pi x)$
- but Fourier says: ← Chapter 4

$$c_n = 2 \int_0^1 \sin(n\pi t) \phi(t) dt$$

- thus, assuming we may exchange the sums:

$$\begin{aligned} u(x) &= \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} \left(\int_0^1 \sin(n\pi t) \phi(t) dt \right) \sin(n\pi x) \\ &= \int_0^1 \left[\sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} \sin(n\pi t) \sin(n\pi x) \right] \phi(t) dt \end{aligned}$$

- guess what! we have the kernel again:

$$k(x, t) = \min\{x, t\} - xt = \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} \sin(n\pi t) \sin(n\pi x)$$

- and again $u(x) = \int_0^1 k(x, t) \phi(t) dt$

to define:

- open and closed sets in metric spaces
- compact sets in metric spaces
- equicontinuity of subsets of $C([a, b])$
- separable metric spaces
- complete metric spaces
- Hilbert and Banach spaces

to prove:

- Heine-Borel theorem
- Arzela-Ascoli theorem
- $C([a, b])$ with $\| \cdot \|_{\infty}$ is complete (thus a Banach space)
- Assignment 2 is posted at bueler.github.io/fa